4	**(a)**	p.d. = $\frac{\text{energy transformed from electrical to other forms}}{\text{unit charge}}$	B1	
		e.m.f. = $\frac{\text{energy transformed from other forms to electrical}}{\text{unit charge}}$	B1	[2]
	(b) (i)	sum of e.m.f.s (in a closed circuit) = sum of potential differences	B1	[1]
	(ii)	$4.4 - 2.1 = I \times (1.8 + 5.5 + 2.3)$	M1	
		$I = 0.24$ A	A1	[2]
	(iii)	arrow (labelled) I shown anticlockwise	A1	[1]
	(iv)	**1.** $V = I \times R = 0.24 \times 5.5 = 1.3(2)$ V	A1	[1]
		2. $V_A = 4.4 - (I \times 2.3) = 3.8(5)$ V	A1	[1]
		3. *either* $V_B = 2.1 + (I \times 1.8)$ *or* $V_B = 3.8 - 1.3$	C1	
		$= 2.5(3)$ V	A1	[2]

5 (a) transverse waves have vibrations that are perpendicular / normal